Pi0.5-LIBERO Reproduction Report

Single RTX 4090D / LeRobot / LIBERO Reproduction Status
Repository: https://github.com/cosmorithm/pi05-libero-reproduction

Executive Summary

The project reproduces the Pi0.5-LIBERO evaluation path on a single NVIDIA RTX 4090 D (24 GB). Simulation, asset installation, two-camera policy input, the Pi0.5 checkpoint, and the gated PaliGemma tokenizer are installed and verified. The first real closed-loop evaluation succeeded. A web dashboard now runs the real Pi0.5 policy while recording all seven camera views. A reproducible four-suite benchmark runner is currently executing.

As of 2026-10-17 05:55 CST, the `libero_spatial` baseline was approximately 73/100 episodes complete. This report documents the protocol and current state rather than presenting a final benchmark success rate.

Environment

Item	Value
Server host	autodl-container-470e41955c-15394eb1
GPU	NVIDIA GeForce RTX 4090 D, 24564 MiB
Disk	50 GB total, about 30 GB free on /root/autodl-tmp
Python	3.12.3
PyTorch	2.11.0+cu128
torchvision	0.26.0+cu128
LeRobot	0.6.2
LIBERO	hf-libero 0.1.4, robosuite 1.4.0, MuJoCo 3.8.1
Checkpoint	lerobot/pi05_libero_finetuned
Checkpoint revision	7152f13e34c6c115d808aea0e5d08aca74eac14b
Tokenizer	google/paligemma-3b-pt-224, GemmaTokenizer, 257152 vocab

Completed Components

- Headless LIBERO simulation: EGL rendering, reset/step, 7 camera views at 256x256.
- Web dashboard v3: idle/demo/Pi0.5 modes, task language, progress, GPU telemetry, token redaction.
- Real Pi0.5 dashboard mode: official processors, n_action_steps=10, seven-view H.264 recording.
- Single-task closed loop: libero_spatial task 0, seed 7, success=true.
- Benchmark smoke: libero_spatial task 0, seed 7, 1/1 success through benchmark runner.
- Resumable benchmark runner and overnight four-suite orchestration script.
- Server-local Git on main, pushed to GitHub with generated artifacts and credentials excluded.

Evaluation Protocol

Field	Value
Policy	lerobot/pi05_libero_finetuned

Field	Value
Suites	libero_spatial, libero_object, libero_goal, libero_10
Tasks per suite	10
Episodes per task	10
Seed	1000
Batch size	1
Action chunk executed	n_action_steps=10
Compile	false
Control mode	relative
Observation	2 x 256x256 RGB + 8D state
Reset	init_states=true, hard_reset=true
Video	official lerobot-eval agentview MP4

Smoke Result

Metric	Value
Suite / task / seed	libero_spatial / 0 / 7
Episodes	1
Success	1/1, success=true
Rollout time	8.12 seconds
Observed GPU memory	about 9.87 GiB
Output	outputs/benchmark/pi05-libero-smoke

Current Full Baseline Progress

The following snapshot was collected at 2026-10-17 05:55 CST while the full `libero_spatial` run was still active. Video files are written after each episode and are used here only as a progress indicator; final success rates come from eval_info.json.

Task	Episodes recorded
0	10
1	10
2	10
3	10
4	10
5	10
6	10
7	3 or more, still running
8	not started
9	not started
Total snapshot	approximately 73/100

Outputs and Reproducibility

Benchmark outputs are stored under `outputs/benchmark/pi05-libero/`. Each suite contains `command.txt`, `metadata.json`, `run.log`, `eval_info.json`, and videos. Top-level `summary.json`, `summary.csv`, and `summary.md` are generated after all requested suites finish.

Reproduce the smoke test:

```
./.venv/bin/python scripts/benchmark_libero.py --suites libero_spatial --task-ids 0 --n-episodes 1 --seed 7 --output-root outputs/benchmark/pi05-libero-smoke
```

Run one suite:

```
./.venv/bin/python scripts/benchmark_libero.py --suites libero_spatial --n-episodes 10 --seed 1000 --output-root outputs/benchmark/pi05-libero
```

Run all four suites:

```
./.venv/bin/python scripts/benchmark_libero.py --suites libero_spatial,libero_object,libero_goal,libero_10 --n-episodes 10 --seed 1000 --output-root outputs/benchmark/pi05-libero
```

Git and Security

The project repository is tracked in server-local Git on `main` and pushed to https://github.com/cosmorithm/pi05-libero-reproduction. Commit history, scripts, and documentation are public. Downloaded weights, caches, assets, logs, outputs, virtual environment, dashboard tokens, and credentials are excluded by .gitignore.

The most recent code commit included in this report is `9272d18 Document benchmark protocol and add overnight runner`.

Next Steps

- Finish libero_spatial, then automatically continue libero_object, libero_goal, and libero_10.
- Compare the four-suite success rates against the published LeRobot reference: Spatial 97.0%, Object 99.0%, Goal 98.0%, Long 96.0%.
- Add more seeds after the one-seed baseline if runtime and GPU budget permit.
- Add seven-view video playback/download to the dashboard and consider a persistent Pi0.5 worker.
- Download LIBERO training data and validate 24 GB fine-tuning only after the baseline is trusted.

Limitations

This report is a live status snapshot. The current benchmark is running on one GPU, one policy checkpoint, one seed, and batch size 1. Final benchmark conclusions must be read from the generated summary files after the overnight run completes.